WCN1-XX56XX-XXXR
WCN2-XX56XX-XXXR
WCN3-XX56XX-XXXR
WCN4-XX56XX-XXXR

SERIES

0.56"SEVEN-SEGMENT NUMERIC LED DISPLAYS

FEATURES

- High intensity and reliability.
- High quality and low cost.
- Choice of colors: Red/Orange/Green/Blue,etc
- Low power requirement.
- I. C. compatible.
- Easy assembly.

DESCRIPTION

The WCN1-XX56XX-XXXR,WCN2-XX56XX-XXXR, WCN3-XX56XX-XXXR and WCN4-XX56XX-XXXR series are 0.56 inch (14.2mm) height single, dual, triad and quad digit displays.

SH. Red displays have black face or gray face and milky segment or red segment.

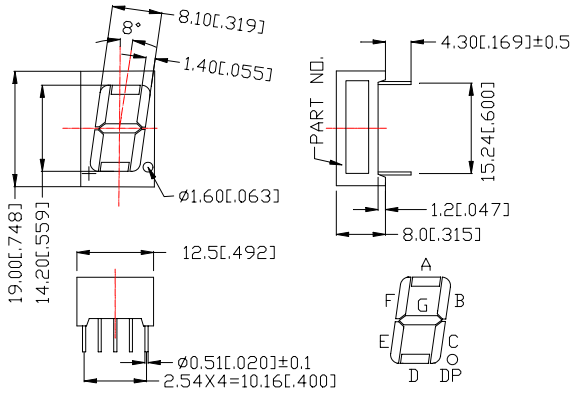
Orange displays have black face or gray face and milky segment or red segment.

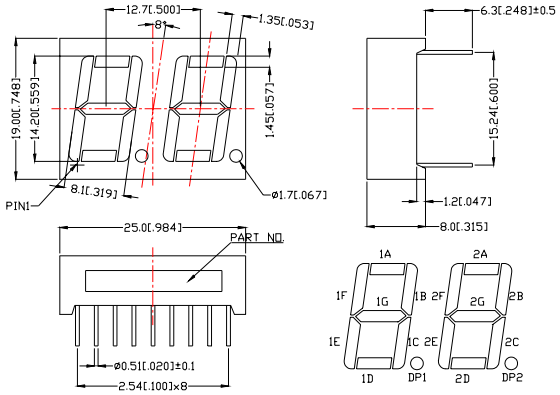
Bright Green displays have black face or gray face and milky segment or green segment.

PACKAGE DIMENSIONS

A. WCN1-XX56XX-A1XR/C1XR

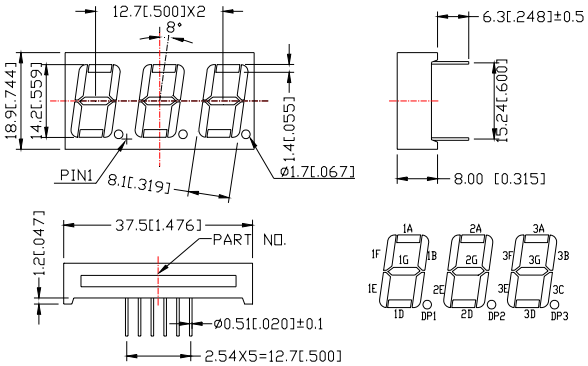
B. WCN2-XX56XX-A1XR/C1XR



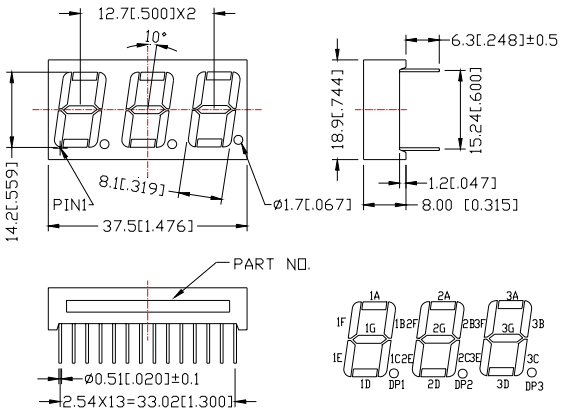


PACKAGE DIMENSIONS

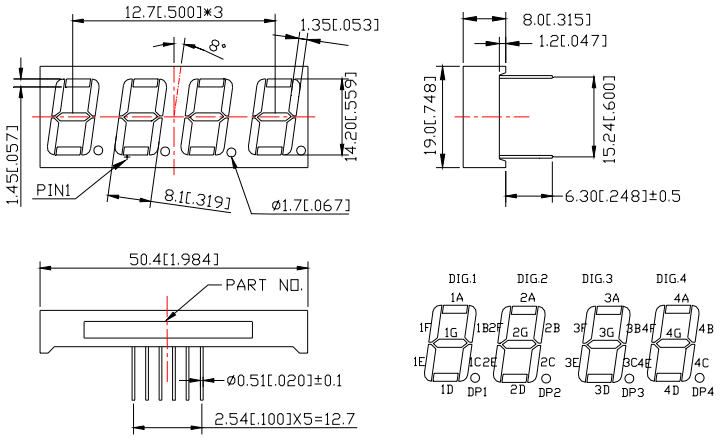
C. WCN3-XX56XX-A1XR/C1XR



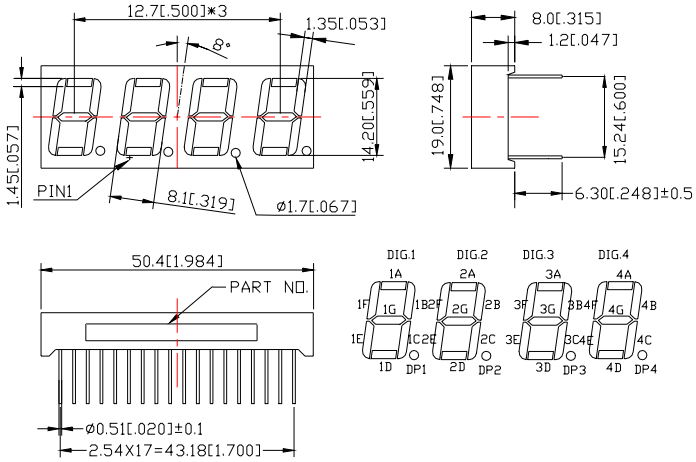
D. WCN3-XX56XX-A2XR/C2XR



E. WCN4- XX56XX-A1XR/C1XR



F. WCN4-XX56XX-A2XR/C2XR



NOTES: All dimensions are in millimeters (inches) tolerance are± 0.25mm(0.010) unless otherwise noted.

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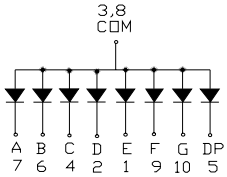
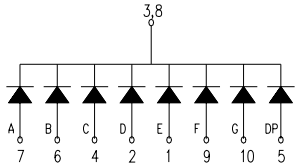
INTERNAL CIRCUIT DIAGRAM

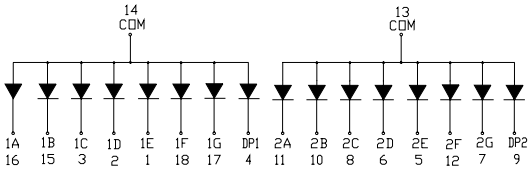
A. WCN1-XX56XX-A1XR



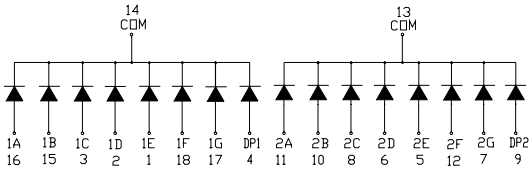
B. WCN1-XX56XX-C1XR



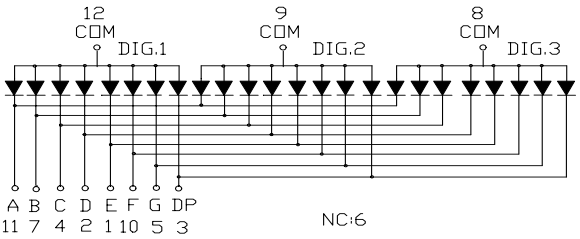
C. WCN2-XX56XX-A1XR



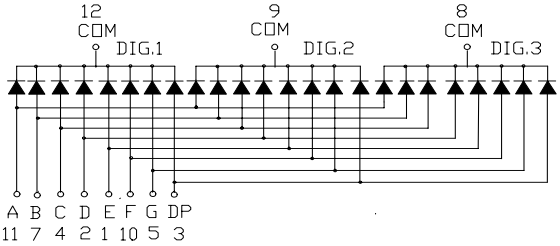
D. WCN2-XX56XX-C1XR



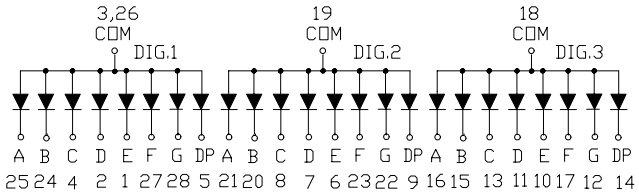
E. WCN3-XX56XX-A1XR



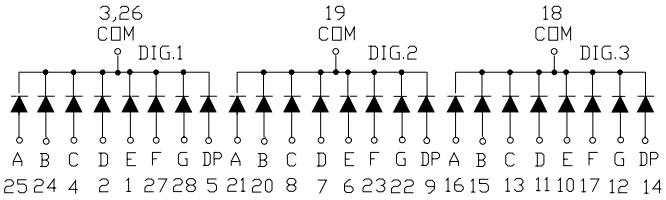
F. WCN3-XX56XX-C1XR



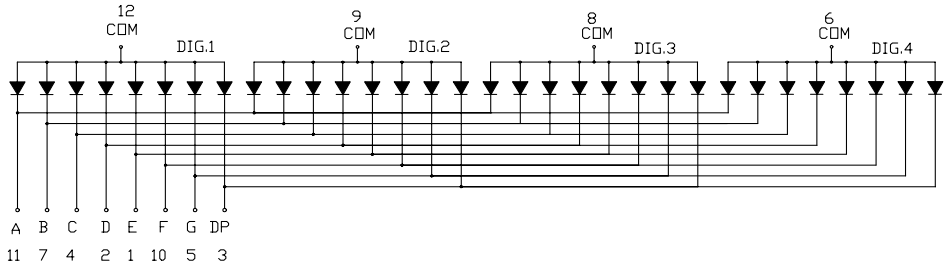
G. WCN3-XX56XX-A2XR



H. WCN3-XX56XX-C2XR

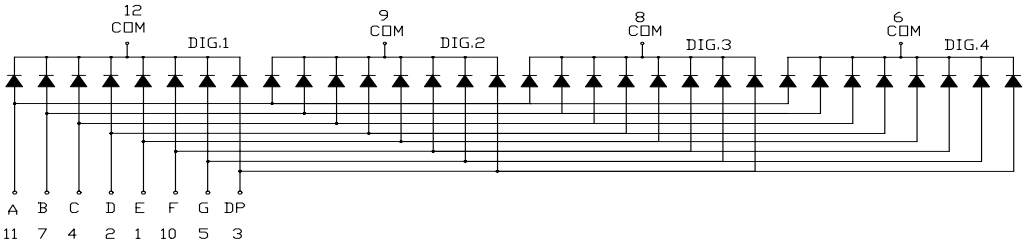


I. WCN4-XX56XX-A1XR

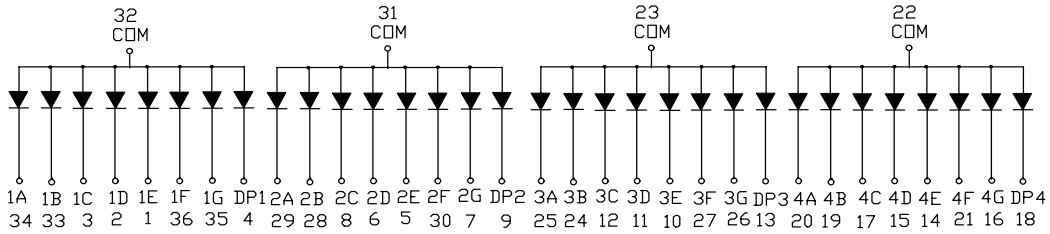


INTERNAL CIRCUIT DIAGRAM

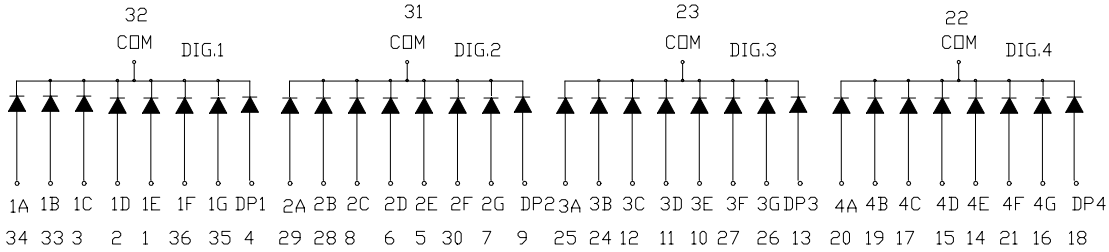
J. WCN4-XX56XX-C1XR



K. WCN4-XX56XX-A2XR



L. WCN4-XX56XX-C2XR



ABSOLUTE MAXIMUM RATINGS AT T_a=25°C

PARAMETER	SH.RED	ORANGE	BRIGHT GREEN	UNIT
Power Dissipation Per Segment	50	65	65	mW
Peak Forward Current Per Segment (1/10duty cycle 0.1ms pulse width)	100	100	100	mA
Continuous Forward Current Per Segment Derating Linear From 25°C Per Segment	25 0.30	25 0.20	25 0.33	mA mA/°C
Reverse Voltage Per Segment	5	5	5	V
Operating Temperature Range	-35°C to + 85°C			
Storage Temperature Range	-35°C to + 85°C			
Solder Temperature 1/16 inch below seating plane for 3 seconds at 260°C				

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ELECTRICAL/OPTICAL CHARACTERISTICS AT T_a=25°C

WCN1-0056SR-A11R/C11R;WCN2-0056SR-A11R/C11R;WCN3-0056SR-A11R/C11R/A21R/C21R
WCN4-0056SR-A11R/C11R/A21R/C21R

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNIT	TEST CONDITION
Luminous Intensity	I_V	3.0	6.0	—	mcd	I_F=10mA
Dominant Wavelength	λ_D	—	643	—	nm	I_F=20mA
Peak Emission Wavelength	λ_P	—	660	—	nm	I_F=20mA
Spectral Line Half-Width	$\Delta\lambda$	—	20	—	nm	I_F=20mA
Forward Voltage Per Segment	V_F	—	1.8	2.0	V	I_F=20mA
Reverse Current Per Segment	I_R	—	—	100	μA	V_R=5V
Luminous Intensity Matching Ratio (Segment To Segment)	I_{V-m}			2:1		I_F=10mA

WCN1-0056HO-A11R/C11R;WCN2-0056HO-A11R/C11R;WCN3-0056HO-A11R/C11R/A21R/C21R

WCN4-0056HO-A11R/C11R/A21R/C21R

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNIT	TEST CONDITION
Luminous Intensity	I_V	1.25	3.5	—	mcd	I_F=10mA
Dominant Wavelength	λ_D	—	622	—	nm	I_F=20mA
Peak Emission Wavelength	λ_P	—	632	—	nm	I_F=20mA
Spectral Line Half-Width	$\Delta\lambda$	—	35	—	nm	I_F=20mA
Forward Voltage Per Segment	V_F	—	2.05	2.6	V	I_F=20mA
Reverse Current Per Segment	I_R	—	—	100	μA	V_R=5V
Luminous Intensity Matching Ratio (Segment To Segment)	I_{V-m}			2:1		I_F=10mA

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNIT	TEST CONDITION
Luminous Intensity	I_V	2.0	4.5	—	mcd	I_F=10mA
Dominant Wavelength	λ_D	—	573	—	nm	I_F=20mA
Peak Emission Wavelength	λ_P	—	568	—	nm	I_F=20mA
Spectral Line Half-Width	$\Delta\lambda$	—	30	—	nm	I_F=20mA
Forward Voltage Per Segment	V_F	—	2.25	2.6	V	I_F=20mA
Reverse Current Per Segment	I_R	—	—	100	μA	V_R=5V
Luminous Intensity Matching Ratio (Segment To Segment)	I_{V-m}			2:1		I_F=10mA